water and is usually clear. It appears as a transparent and colourless fluid. The cytoplasm serves as a molecular soup. It is in the cytoplasm where all the cellular organelles are suspended and are bound together by a lipid bilayer membrane. The cytoskeleton present in the cytoplasm gives the cell its shape. Cytoplasm also constitutes numerous salts and is a very good conductor of electricity.

Various metabolic activities occur in the cytoplasm. Metabolic pathways like glycolysis and cellular processes like cell division take place in the cytoplasm.

- Cytoplasm shows differential staining properties, the areas stained with the basic dyes are the basophilic areas of the cytoplasm and are termed as ergatoplasm for this material.
- It is a heterogenous mixture of opaque granules and organic compounds which gives it its colloidal nature.
- The cytoplasm conatains dissolved nutrients and it aids to dissolve waste products.
- It helps movement of the cellular materials around the cell through a process called cytoplasmic streaming.
- The peripheral zone of cytoplasm is jelly-like and is known as the plasmogel. The surrounding area of the nuclear zone is thin and liquefied in nature and is known as the plasmosol.
- The physical nature of cytoplasm is colloidal. It has a high percentage of water and particles of various shapes and sizes are suspended in it.
- It also contains proteins, of which 20-25 percent are soluble proteins including enzymes.
- Also, certain amount of carbohydrates, RNAs, inorganic salts and lipid substances are found.
- The plasmogel part of the cytoplasm is capable of absorbing water and removing it, according to the cell's need.
- The stomatal guard cells present in the leaves exhibit this property.
- An organized system of fibres can be observed by specific staining techniques.

1.4.11. Plastids

Plastids are found in all plant cells and in euglenoides. These are easily observed under the microscope as they are large. They bear some specific pigments, thus imparting specific colours to the plants. Based on the type of pigments

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